

then fed to port pin PC2 of microcontroller IC1. Capacitor C4 is used for filtering and zener diode ZD1 is used to prevent this voltage from exceed-

ing 5V and burn ADC channel of the microcontroller. The voltage scaling is reconfigurable via preset VR1.

The brain of the circuit, which man-

ages all the input/output and data manipulations, is microcontroller IC1. Pin 21 of IC1 is directly connected to reference input of the ADC unit inside the microcontroller and, because we are using the internal 2.56V reference already built inside the microcontroller, we by-pass this pin to ground using a 0.1µF ceramic capacitor.

Analogue temperature sensor IC2 is used to sense the room temperature. The output pin of IC2 is directly connected to port pin PC0 of IC1 to sense the room temperature and display it on the LCD. For audio alarm indication, a piezo-buzzer with built-in beeping circuit is connected to PB0 of IC1 that produces the alert sound in case of voltage malfunction. To display the output power status, LED2 is connected at port pin PB1 of IC1.

Port pins PD4, PB6, PB7 and PD5 of IC1 are connected to pins D4 through D7 of LCD1, respectively. Port pins PD6 and PD7 are connected to control pins RS and EN while the R/W pin is kept low to always keep it in write mode. Pin 15 of LCD1 is driven through transistor T2, whose base is connected to port pin PC4 of IC1 through resistor R6. This enables us to switch on/off backlight of the LCD from the microcontroller. Port pin PB2 is used to drive a power relay for switching on/off the load.

Port pins PD0 through PD3 of IC1 are connected with tactile switches S1 through S4, respectively, to let the user change the settings of the device and to set the under and over voltage cut-off limits of mains input voltage.

Once all the connections are made, switch on the power supply and you will see a welcome message on the LCD for one or two seconds and then the display will revert back to displaying the voltage and temperature as shown in Fig. 4.

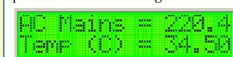


Fig. 4: Screen display

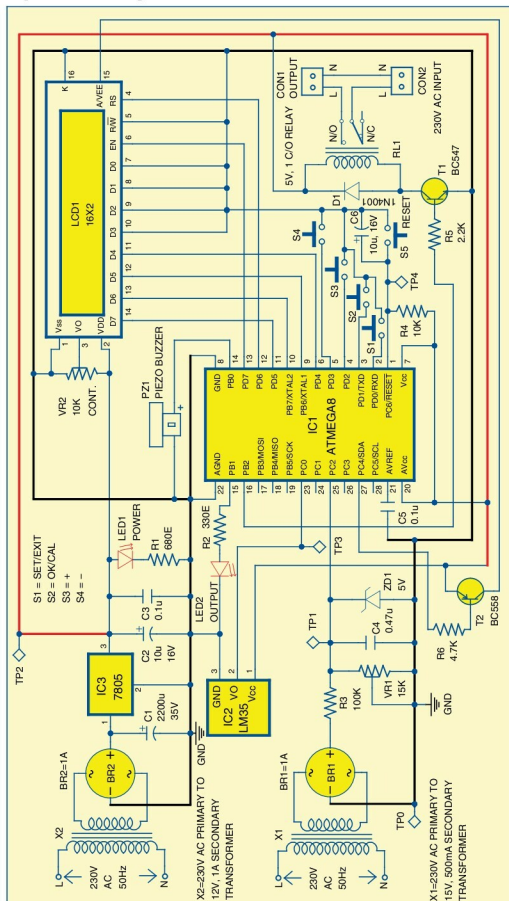


Fig. 3: Circuit diagram of safety timer for home appliances